ISS Tester and Trainer Handbook

This document is the zero-knowledge onboarding guide for manual testers, business users, and trainers working with the ISS ERP system.

It combines the "what is this system?", "how do I run it?", and "how do I test it?" answers into one place.

Validated against the current repository and a live local walkthrough on March 14, 2026.

1. What This System Is

ISS is a browser-based ERP system for service, inventory, procurement, sales, finance, and reporting work.

Main business areas:

- Master data
- Procurement
- Inventory
- Sales
- Service
- Finance
- Reporting
- Admin and audit

The system is built with:

- Frontend: Next.js web app
- Backend: ASP.NET Core Web API
- Database: PostgreSQL
- Supporting services: PDF generation, document attachments, audit logging, notifications

2. Who Should Use This Guide

Use this guide if you are:

- a new tester who has never used an ERP before

- a trainer preparing a demo or user onboarding session
- a business user validating that a change still works
- a project lead who needs a repeatable UAT script

Use the deeper technical docs only when you need implementation detail:

- `docs/manual-uat-guide.md`
- `docs/user-manual.md`
- `docs/system-technical-maintainer-guide.md`
- `docs/backend-architecture.md`
- `docs/frontend-architecture.md`

3. Basic ERP Concepts

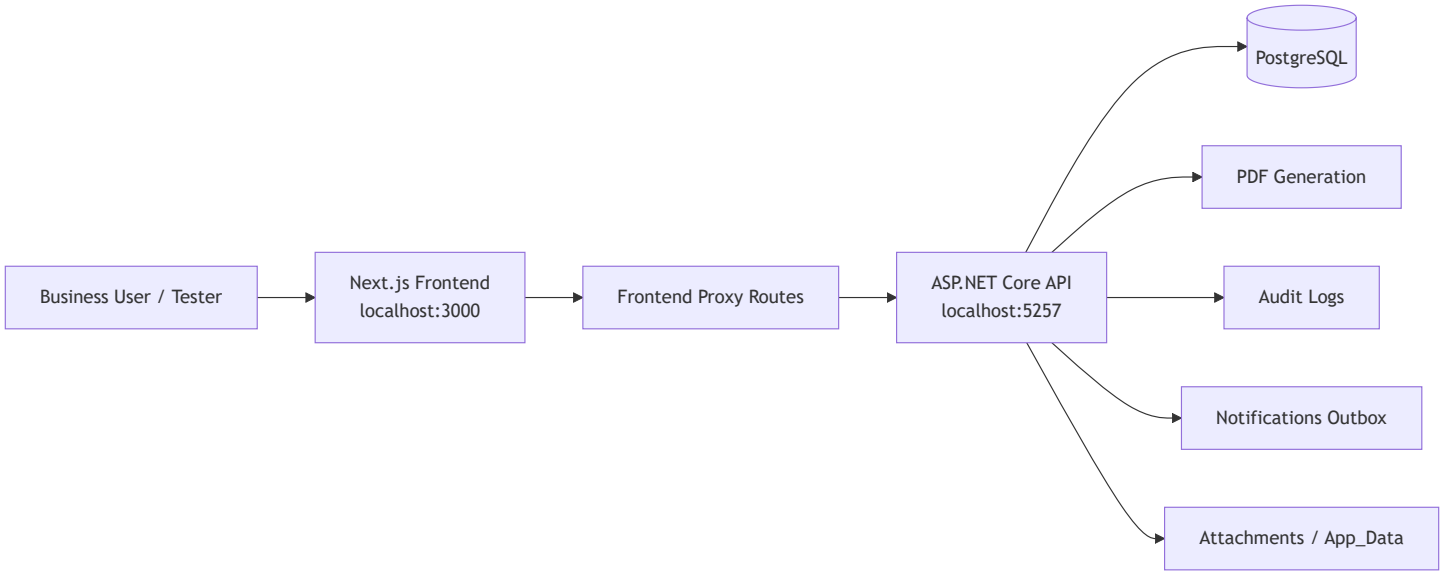
If the trainee is completely new, teach these first:

Term	Meaning in ISS
Master data	Reference data used by transactions, such as items, suppliers, customers, warehouses, taxes, currencies
Transaction	A business document such as a PO, GRN, invoice, payment, stock adjustment, or work order
Draft	A document that can still be edited
Posted	A document that has created business impact, such as stock movement or AR/AP
AR	Accounts receivable, meaning money customers owe you
AP	Accounts payable, meaning money you owe suppliers
On hand	Current physical stock quantity
Costing	Inventory value based on default and weighted average cost
Audit log	History of changes and actions in the system
PDF output	Downloadable printable version of a business document

Common status meanings:

Status	Meaning
Draft	Being prepared
Approved / Confirmed	Accepted for the next stage
Sent	Shared with the outside party
Posted	Financial or inventory effect applied
Paid	Invoice fully settled
Completed / Closed	Service flow finished
Void / Cancelled	Stopped and no longer active

4. System Overview



What this means in practice:

- users work entirely in the browser
- the frontend sends requests to the backend API
- the backend saves business data in PostgreSQL
- posted transactions update stock, AR/AP, reports, audit history, and PDFs

5. Functional Map

Menu area	Purpose	Examples of what to test
Dashboard	High-level KPIs	AR/AP totals, reorder alerts, service jobs
Master Data	Core setup records	currencies, items, warehouses, suppliers, customers
Procurement	Buying and receiving	PR, RFQ, PO, GRN, supplier invoice, supplier return
Inventory	Stock control	on hand, reorder alerts, stock adjustment, stock transfer
Sales	Selling and dispatching	quotes, orders, dispatches, invoices, customer returns
Service	Workshop and after-sales work	jobs, work orders, estimates, QC, handovers
Finance	Money owed and paid	AR, AP, payments, credit notes, debit notes
Reporting	Operational visibility	stock ledger, aging, tax summary, service KPIs, costing
Admin	Control and support tools	users, notifications, Excel import, settings
Audit Logs	Change tracking	posted actions and entity changes

6. Local Setup For Testing

From the repo root:

Database

```
docker compose up -d
```

Expected:

- PostgreSQL is available on `localhost:5433`

Backend

```
dotnet run --project backend/src/ISS.Api/ISS.Api.csproj
```

Expected:

- API starts on `http://localhost:5257`
- `http://localhost:5257/health` returns `Healthy`

Frontend

```
cd frontend
copy .env.example .env.local
npm install
npm run dev
```

Expected:

- app loads on `http://localhost:3000`

Login and roles

- In a fresh system, the first registered user becomes `Admin`
- Existing role names are:
  - `Admin`
  - `Procurement`
  - `Inventory`
  - `Sales`
  - `Service`
  - `Finance`
  - `Reporting`

Trainer note:

- for local demos, always use an account with `Admin` access so every screen is visible

7. Recommended Training Sequence

Train new testers in this order:

1. Explain the module map and the meaning of `Draft`, `Posted`, `AR`, `AP`, and `On Hand`.
2. Show login, sidebar navigation, and the dashboard.
3. Run the smoke test.
4. Run one full end-to-end business scenario.
5. Teach the tester how to collect evidence:
    - screenshot
    - document number
    - expected value
    - actual value
    - page URL
6. Move to the full module regression checklist.

The best first full scenario is:

- create master data
- receive stock through procurement
- sell the stock
- create AR/AP entries
- validate reports

8. Smoke Test

Use this for every build or release candidate.

Smoke steps

1. Open `Master Data -> Currencies`.
2. Open `Finance -> Payments`.
3. Open `Reporting -> Costing`.
4. Open `Admin -> Users`.
5. Open `Audit Logs`.

Expected results

- currencies page loads and base currency exists
- payments page loads without currency errors
- costing page loads successfully
- users page loads for admin users
- audit log page opens without errors

If these five fail, stop deeper testing and fix the environment first.

9. Core End-To-End Scenario

This is the main business walkthrough a new tester should learn first.

It follows a simple stock-in -> sell -> finance -> reporting path.

Scenario summary

Step	Business meaning	Expected business effect
Create master data	Create the records the system depends on	Records are saved and selectable
Create PO and GRN	Buy and receive stock	On hand increases and AP is created
Create direct dispatch and invoice	Sell stock to a customer	On hand decreases and AR is created
Review finance pages	Confirm money owed	AR and AP show correct balances
Review costing	Confirm stock valuation	On hand and value match transaction history

Suggested test values

Use the same values every time so results are easy to compare.

Type	Value
Warehouse	`MAIN`
Supplier	`SUP1`
Customer	`CUS1`
Item	`SKU1 - Hydraulic Filter`
Receipt quantity	`10`

Type	Value
Receipt unit cost	`5`
Sales quantity	`4`
Sales unit price	`7`

Step A: Create core master data

Create:

- one warehouse
- one supplier
- one customer
- one item with UoM `PCS` and default cost `5`

Expected:

- each record saves without error
- each record appears in its list page

Evidence to capture:

- screenshot of the saved row
- code or SKU used

Step B: Procurement receipt

1. Create a purchase order for the supplier.
2. Add one line for the item:
    - quantity `10`
    - unit price `5`
3. Approve the PO.
4. Create a goods receipt from the PO.
5. Add one GRN line:
    - quantity `10`
    - unit cost `5`
6. Post the GRN.

Expected:

- PO total is `50`
- GRN posts successfully
- AP shows an outstanding supplier balance of `50`
- On Hand becomes `10`

Step C: Sales issue and invoice

1. Create a direct dispatch for the customer and warehouse.
2. Add one line:
    - quantity `4`
3. Post the direct dispatch.
4. Create a sales invoice for the same customer.
5. Add one line:
    - quantity `4`
    - unit price `7`
    - discount `0`
    - tax `0`
6. Post the invoice.

Expected:

- stock drops from `10` to `6`
- invoice total is `28`
- AR shows an outstanding customer balance of `28`

Step D: Reporting validation

Open:

- `Finance -> AP`
- `Finance -> AR`
- `Reporting -> Costing`

Expected:

- AP outstanding = `50`
- AR outstanding = `28`
- costing shows:
    - on hand = `6`
    - default cost = `5`

    - weighted average cost = `5`
    - inventory value = `30`

Step E: Optional payment test

If time allows, continue:

1. Create an incoming payment for the customer for `28`
2. Allocate the payment to the invoice

Expected:

- payment remaining becomes `0`
- AR outstanding becomes `0`
- invoice status changes to `Paid`

10. Full Regression Checklist

Use this after the smoke test and core scenario.

Navigation and security

- login works with valid credentials
- invalid login shows a clear error
- sign out returns to login
- non-admin users only see pages allowed by their role

Master data

- create, edit, save, cancel, and delete work on list pages
- blocked deletes show a useful message when a record is in use
- inactive records behave correctly in dropdowns and lists
- seeded reference data exists on a clean system:
    - currencies
    - payment types
    - tax codes
    - reference forms

Procurement

- RFQ create -> add lines -> send
- purchase requisition create -> submit -> approve
- approved PR converts to PO
- PO create -> add lines -> approve
- GRN create from PO -> add lines -> post
- direct purchase posts stock correctly
- supplier invoice posts AP correctly
- supplier return reduces stock and adjusts supplier balance
- PDFs open and document numbers match the screen

Inventory

- on hand query returns the expected quantity
- stock adjustment draft lines can be add/edit/delete before posting
- posting a stock adjustment changes on hand
- stock transfer create/post works across warehouses
- reorder alerts reflect reorder settings
- reorder alerts can create a PR draft

Sales

- quote create -> add lines -> send
- sales order create -> add lines -> confirm
- dispatch create -> add lines -> post
- direct dispatch posts stock correctly
- invoice post creates AR correctly
- customer return increases stock and adjusts customer balance
- PDFs open and totals match the screen

Service

- equipment unit can be registered
- service job can be created and status changed
- work order can be created
- estimate can be created, lined, and approved or sent
- material requisition can be posted
- quality check can be recorded

- handover can be completed and converted to invoice

Finance

- AR page shows open invoice balances
- AP page shows open supplier balances
- payment create and allocation work
- credit note and debit note flows work
- payment PDF opens and matches the screen

Reporting

- dashboard shows current KPI totals
- stock ledger reflects inventory movements
- aging groups balances correctly
- tax summary loads and shows totals
- service KPI page loads
- costing matches item movement history

Admin and support

- admin can create users and assign roles
- password reset works
- notifications page loads and retry action works when needed
- Excel import template downloads and upload validates correctly
- audit logs show business activity after posting

11. Evidence Rules For Testers

Every defect report should include:

- module and page
- exact steps
- sample data used
- expected result
- actual result
- one screenshot
- document number or record code
- date and time

Recommended defect template:

```
Title:
Module/Page:
Environment:
Role used:
Test data:
Steps to reproduce:
Expected result:
Actual result:
Evidence:
Severity:
```

12. Trainer Guidance

When training a new tester:

- explain the business purpose before the button clicks
- ask the tester what should change after posting a document
- make them predict AR/AP/on-hand before opening the report page
- keep one repeatable dataset per training session
- teach them to verify both the transaction page and the downstream effect

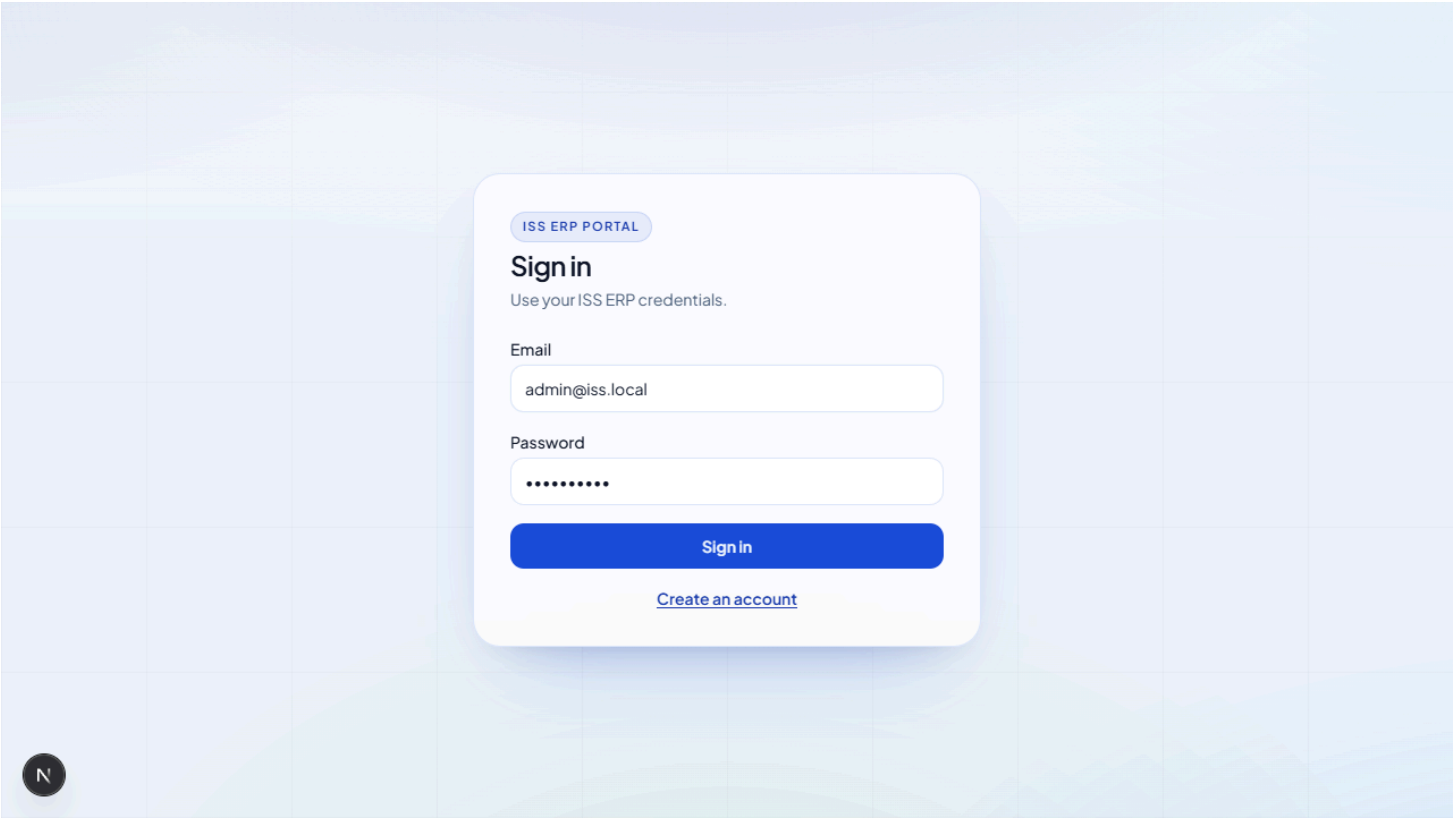
A trainee is ready to test independently when they can explain:

- why master data is needed before transactions
- what posting changes
- how procurement affects stock and AP
- how sales affects stock and AR
- how reports prove the transactions were applied correctly

13. Screenshots

These screenshots were captured from a live local walkthrough on March 14, 2026.

Figure 1: Login screen



The login page also supports account creation when self-registration is enabled.

Figure 2: Dashboard after sample walkthrough

ISS ERP
Service + Inventory + Sales

admin@iss.local
Admin

Settings | Sign out

OVERVIEW

- Dashboard

MASTER DATA

- Brands
- UoMs
- UoM Conversions
- Item Categories
- Items
- Customers
- Suppliers
- Warehouses
- Reorder Settings
- Payment Types
- Tax Codes
- Tax Conversions
- Currencies
- Currency Rates
- Reference Forms

PROCUREMENT

- Purchase Reqs
- RFQs
- Purchase Orders
- Goods Receipts
- Direct Purchases
- Supplier Invoices
- Supplier Returns

SALES

- Quotes
- Orders
- Dispatches
- Direct Dispatches
- Invoices
- Customer Returns

SERVICE

- Equipment Units
- Jobs
- Estimates
- Work Orders
- Material Reqs
- Quality Checks
- Handovers

INVENTORY

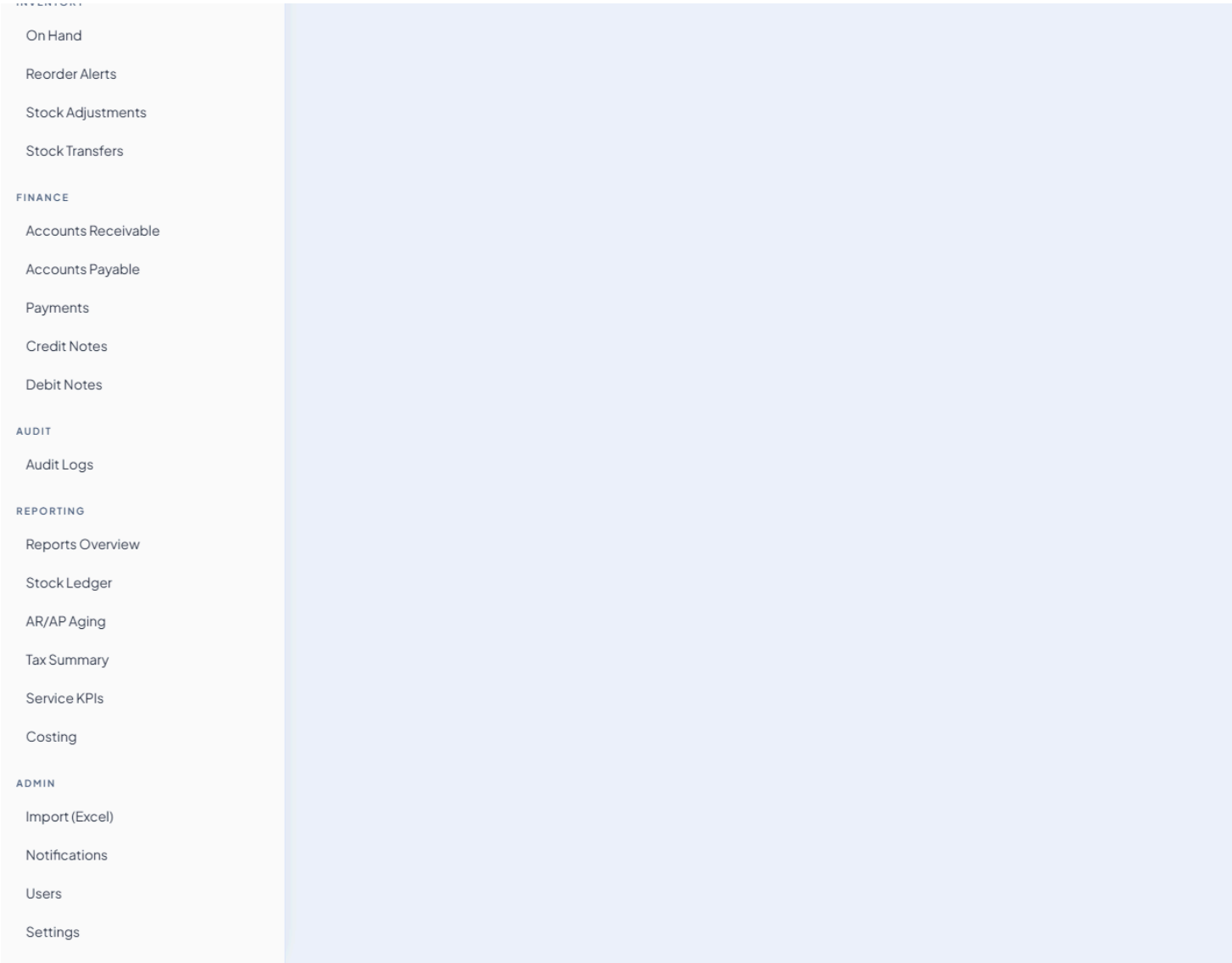
Dashboard

Real-time snapshot across service, finance, and inventory.

Open Service Jobs	AR Outstanding	AP Outstanding	Reorder Alerts
0	LKR 28.00	LKR 50.00	0

Quick actions

New Purchase Order | New Sales Order | Stock Adjustment | New Service Job



In the captured example, the dashboard shows:

- AR outstanding `LKR 28.00`
- AP outstanding `LKR 50.00`
- reorder alerts `0`

Figure 3: Seeded currency master data

OVERVIEW
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INVENTORY

Currencies

Currency master for transactions and costing valuation.

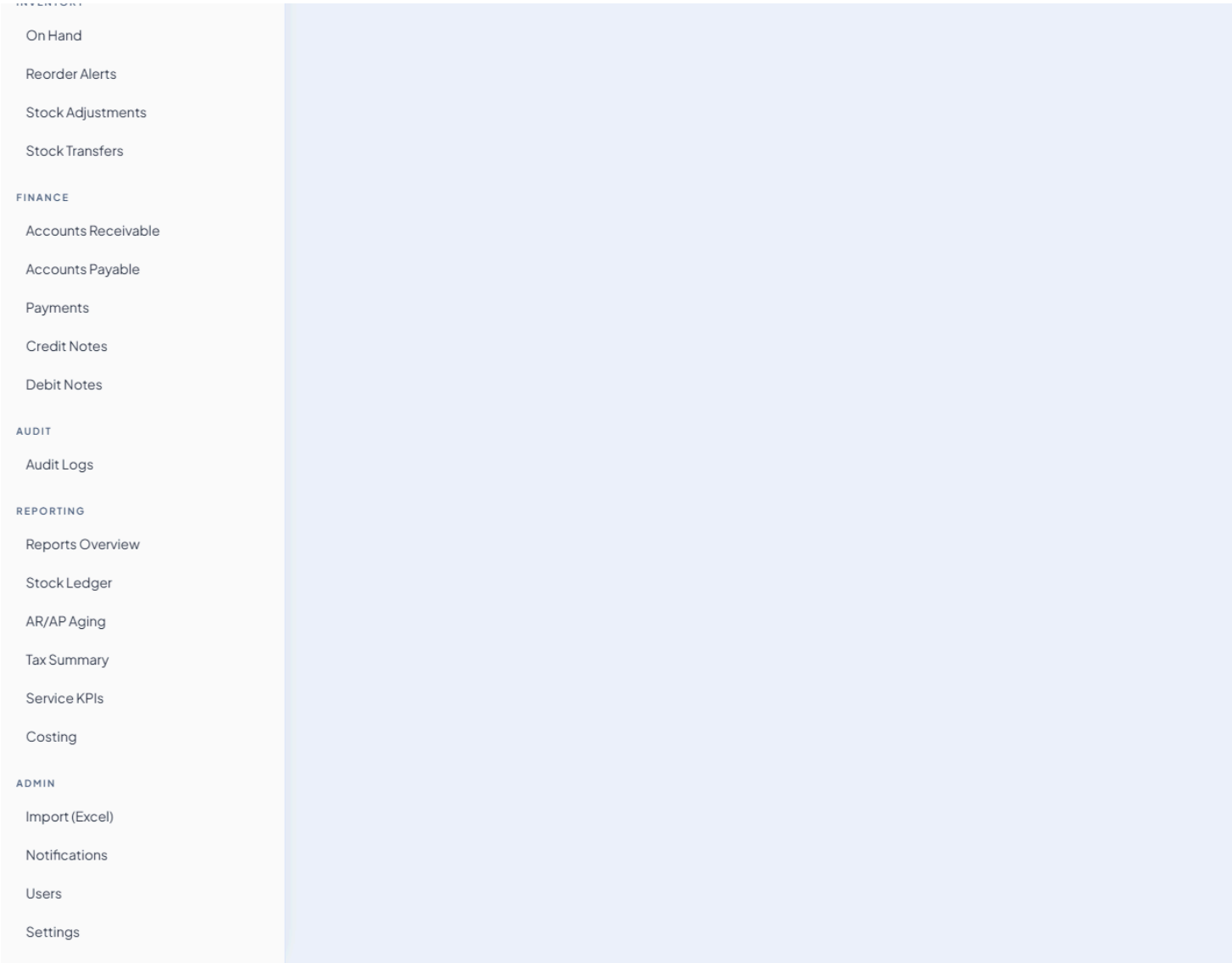
Create

Code: USD
Name: US Dollar
Symbol: $
Minor Units: 2
Base Currency: No

Create Currency

List

CODE	NAME	SYMBOL	MINOR	BASE	ACTIVE	ACTIONS	
EUR	Euro	EUR	2	No	Yes	Edit	Delete
GBP	Pound Sterling	GBP	2	No	Yes	Edit	Delete
USD	US Dollar	$	2	Yes	Yes	Edit	Delete



This is a good smoke-test page because fresh systems should already have default currencies.

Figure 4: Purchase order list and create area

OVERVIEW
- Dashboard

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INVENTORY

Purchase Orders

PO workflow: draft → approve → receive.

Create

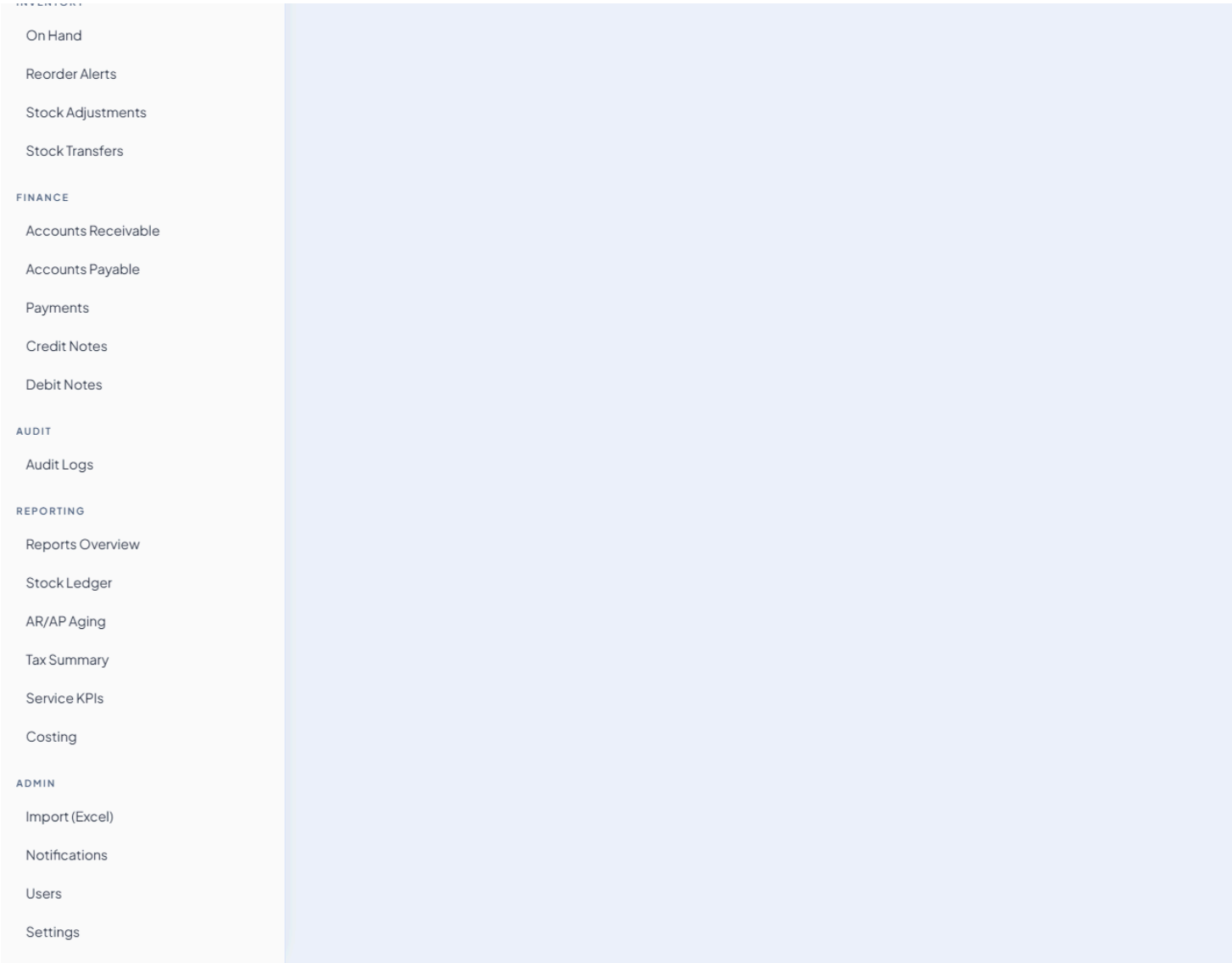
Supplier

Select…

Create Purchase Order

List

NUMBER	SUPPLIER	ORDER DATE	STATUS	TOTAL
P0000003	SUP231252	3/14/2026, 11:12:53 PM	Closed	50
P0000002	sup1	2/26/2026, 8:14:39 PM	Approved	34
P0000001	sup1	2/26/2026, 8:13:37 PM	Draft	0



Use this screen to teach the difference between a draft transaction and an approved document.

Figure 5: Accounts payable showing GRN-driven liability

OVERVIEW
Dashboard

MASTER DATA
Brands
UoMs
UoM Conversions
Item Categories
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Customers
Suppliers
Warehouses
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Invoices
Customer Returns

SERVICE
Equipment Units
Jobs
Estimates
Work Orders
Material Reqs
Quality Checks
Handovers

INVENTORY

Accounts Payable

Outstanding supplier balances (typically from posted GRNs).

Filter

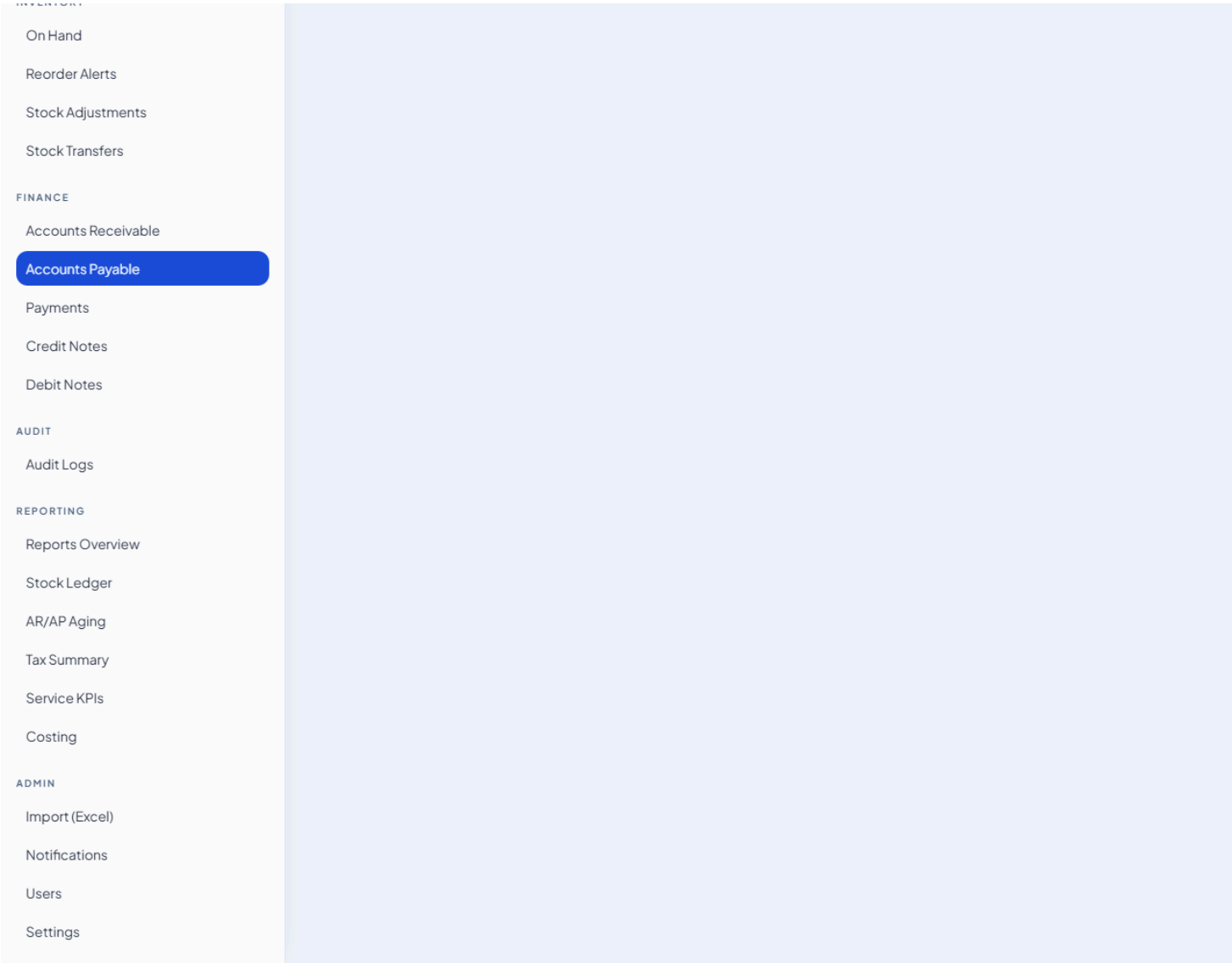
Show

Outstanding only

Apply

Entries

POSTED	SUPPLIER	REF	AMOUNT	OUTSTANDING
3/14/2026, 11:12:53 PM	SUP231252	GRN:2e42eb4a	50	50



In the captured example, supplier `SUP231252` has an outstanding GRN balance of `50`.

Figure 6: Accounts receivable showing invoice-driven balance

OVERVIEW
Dashboard

MASTER DATA
Brands
UoMs
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Handovers

INVENTORY

Accounts Receivable

Outstanding customer balances (typically from posted invoices).

Filter

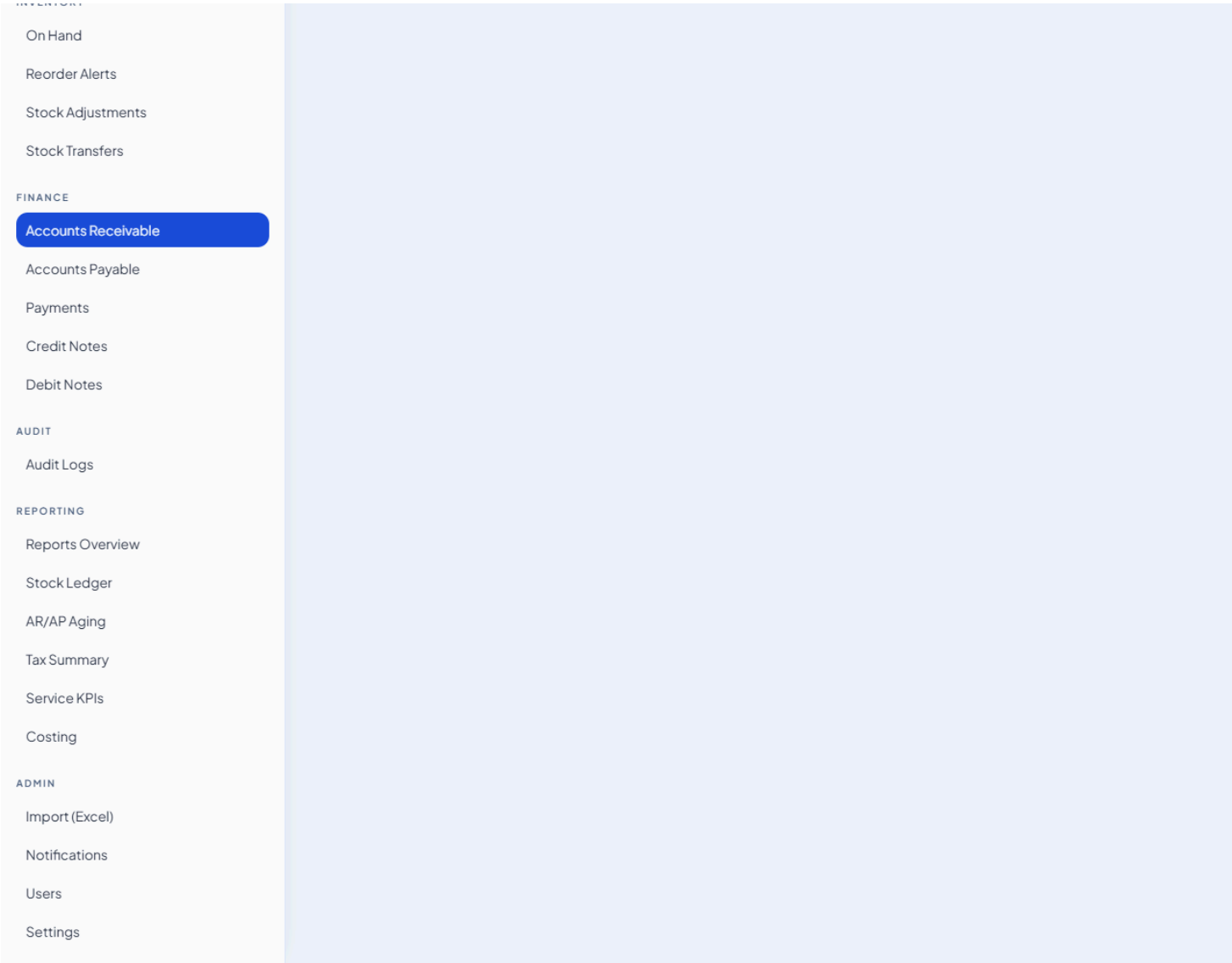
Show

Outstanding only

Apply

Entries

POSTED	CUSTOMER	REF	AMOUNT	OUTSTANDING
3/14/2026, 11:12:54 PM	CUS231252	INV:b5e579f4	28	28



In the captured example, customer `CUS231252` has an outstanding invoice balance of `28` .

Figure 7: Costing report showing valuation after stock in and

stock out

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INVENTORY

Costing

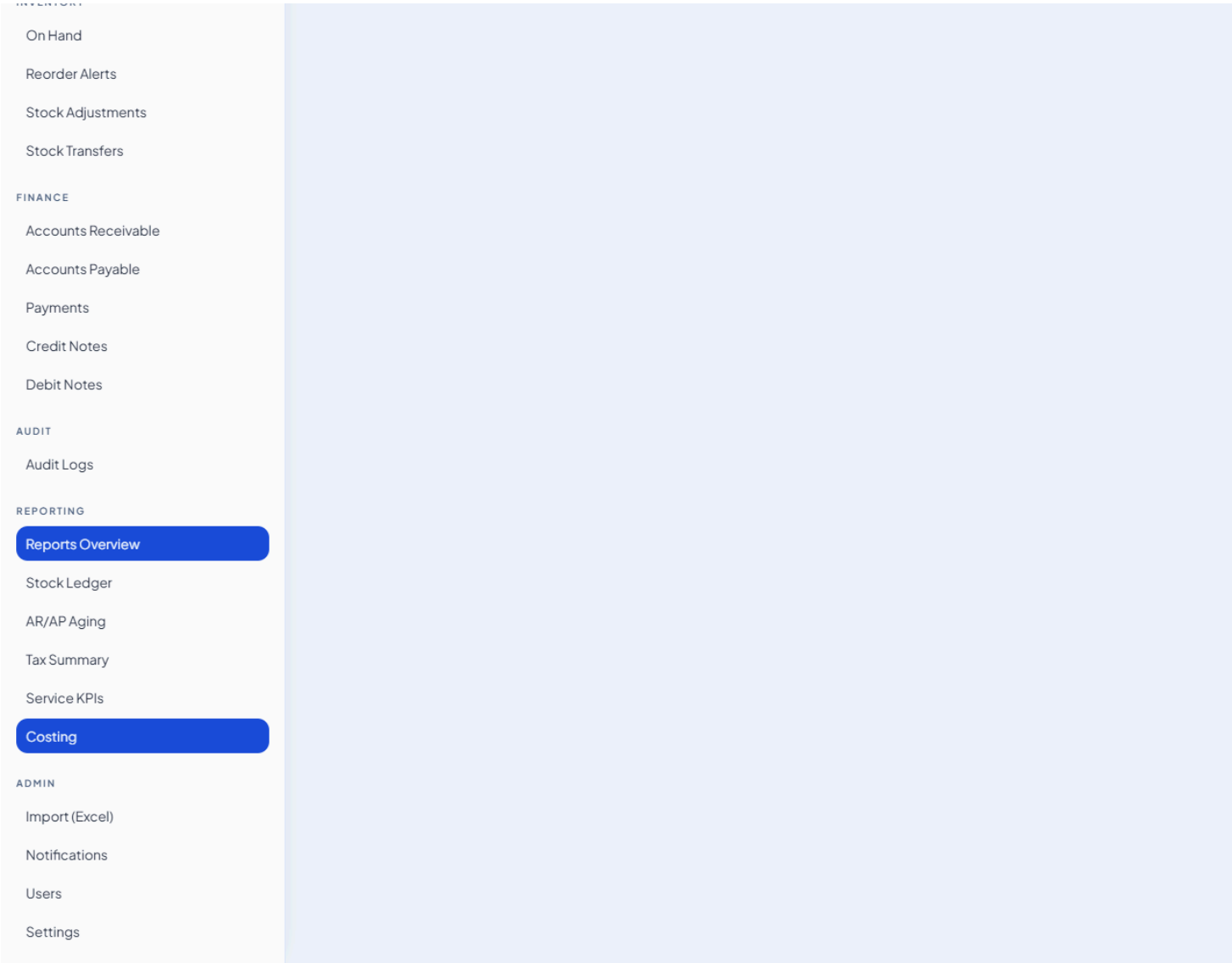
Default vs weighted costs, last receipt rates, and on-hand valuation.

Filter

Warehouse: All warehouses
Item: All items
Take: 500
Apply

ITEMS	TOTAL ON HAND	INVENTORY VALUE (USD)
2	6	$30.00

ITEM	UOM	ON HAND	DEFAULT COST	WEIGHTED AVG COST	LAST RECEIPT COST	LAST RECEIPT AT	VARIANCE %	INVENTORY VALUE
213 2312	PCS	0	$23.00	-	-	-	-	$0.00
SKU231252 Hydraulic Filter	PCS	6	$5.00	$5.00	$5.00	3/14/2026, 11:12:53 PM	0.00%	$30.00



In the captured example, item `SKU231252` shows:

- on hand `6`
- weighted average cost `5`
- inventory value `30`

Figure 8: Admin user management

OVERVIEW
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INVENTORY

Admin · Users

Create users, assign roles, and manage access.

Create

Email | Password | Display name

Roles

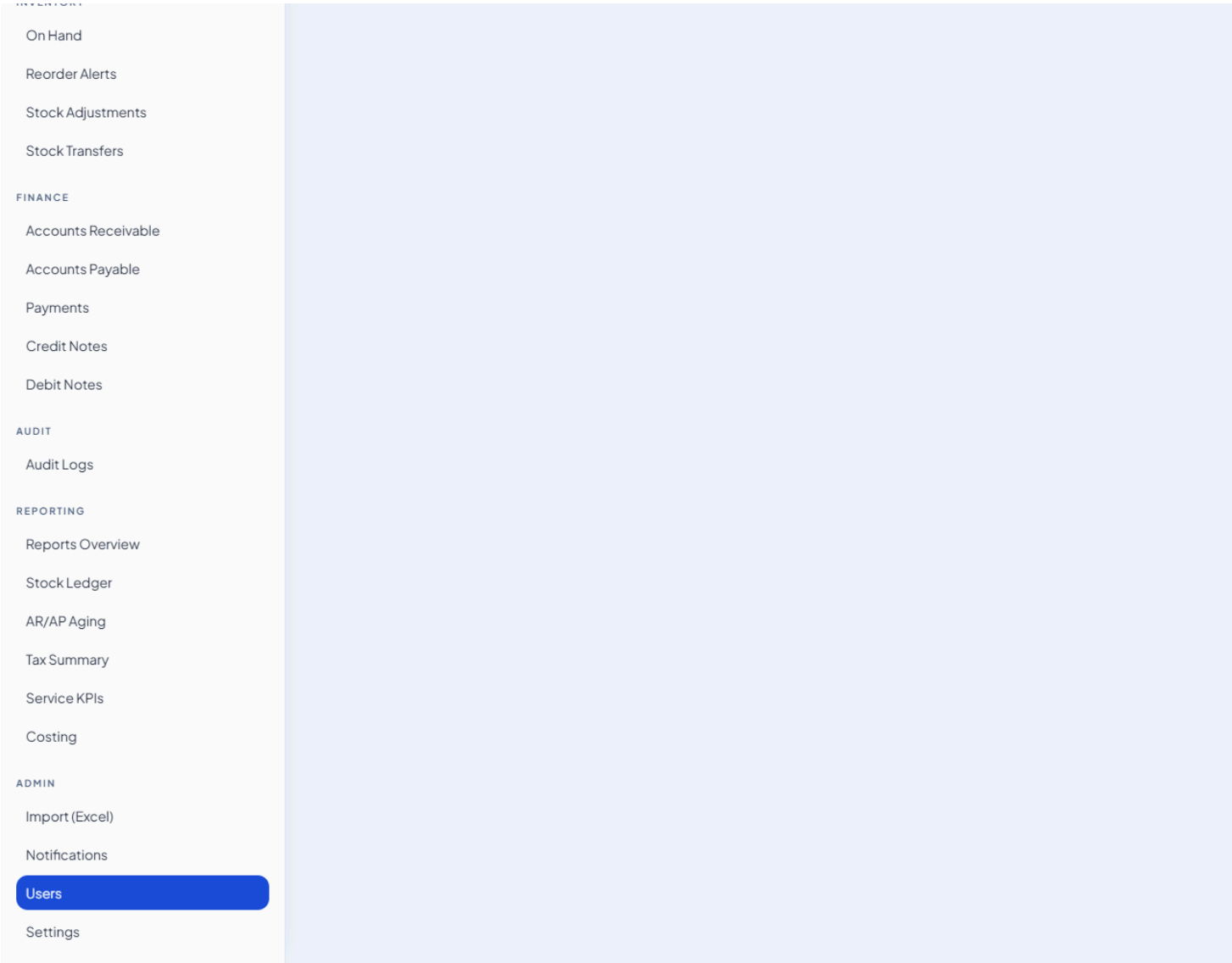
☐ Admin ☐ Procurement ☐ Inventory ☐ Sales ☐ Service ☐ Finance ☐ Reporting

Leave empty for a user with no module access.

Create user

List

EMAIL	NAME	ROLES	STATUS	ACTIONS
admin@iss.local	ISS Admin	Admin	Active	Edit roles · Reset password · Disable
admin@local	admin	Admin	Active	Edit roles · Reset password · Disable



Use this screen to train admins on role assignment and account support tasks.

14. Known Gaps And Current Risks

The system is broad, but the repository already records a few areas that still need careful manual verification:

- full responsive QA across every page is not finished yet
- some proposal-level advanced workflows are still partial:
  - RFQ compare and award depth
  - multi-stage stock transfer depth
  - some richer master-data fields
  - some advanced reporting variants
- "all documents support comments, attachments, PDFs, and logs" still needs page-by-page confirmation

This means testers should pay extra attention to:

- mobile and tablet layouts
- less common workflow branches
- line editing and delete behavior in draft documents
- document PDFs and attachments
- page-to-page consistency

15. Best Supporting Documents

Use this handbook first, then move to these documents when needed:

- `README.md` for local startup and test commands
- `docs/manual-uat-guide.md` for the shorter core UAT path
- `docs/user-manual.md` for module-by-module functional notes
- `docs/system-technical-maintainer-guide.md` for system navigation
- `docs/backend-architecture.md` for backend behavior and operations
- `docs/frontend-architecture.md` for frontend behavior and UI structure